Lesson 33: Normal-Form Games and Pure Nash Equilibrium

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 References **2**

3 Games as Models of Strategic Interaction **3**

4 Normal-Form Games **3**

5 Dominance and Iterated Elimination **5**

6 Best Response and Rationalizability **7**

7 Pure Strategy Nash Equilibrium **9**

8 Analysis of Classic Games **10**

8.1 Prisoner's Dilemma 10

8.2 Battle of the Sexes 10

8.3 Matching Pennies 10

8.4 Stag Hunt 11

9 Exercises **11**

10 Summary **12**

11 Further Reading **13**

1 Motivation and Overview

Consider two autonomous ride-sharing companies operating in the same city. Each company must set a price for its rides—high or low—without knowing the competitor's choice in advance. If both set high prices, both earn comfortable profits. If one undercuts the other, the cheaper service captures most of the market. If both set low prices, both barely break even. How should a company reason about pricing?

This scenario is fundamentally different from single-agent optimization. In a standard optimization problem, there is one decision-maker and the outcome depends only on that decision. Here, each company's profit depends on what the *other* company does. This interdependence of decisions is the defining feature of a *game* in the mathematical sense, and the mathematical framework for analyzing such situations is *game theory*.

RL/ML Connection

In single-agent reinforcement learning, the environment is modeled as a Markov decision process whose transition kernel does not *adapt* to the agent's behavior. Multi-agent RL (MARL) breaks this assumption: each agent's effective "environment" includes the other agents. Every MARL algorithm implicitly targets a game-theoretic solution concept:

- **Independent Q-learning**: each agent ignores others; if it converges, the result is a Nash equilibrium of the stage game.
- **Minimax-Q** [Shoham and Leyton-Brown, 2008]: targets the minimax value in zero-sum stochastic games, applying von Neumann's minimax theorem.
- **Nash-Q**: computes Nash equilibria of stage games at each step.
- **MAPPO, QMIX**: cooperative algorithms whose analysis relies on common-payoff game structure.

Without understanding Nash equilibria and dominance, one cannot state what these algorithms compute, let alone prove convergence.

Prerequisites. This lesson assumes no prior exposure to game theory. The mathematical prerequisites are modest: basic set notation and finite probability spaces (Lesson 9), familiarity with arg max and elementary logic ("for all," "there exists"), and comfort reading tables of numbers. The proof of the IESDS-rationalizability equivalence (Proposition 6.5) uses the supporting hyperplane theorem from convex analysis (Lesson 17), but this result can be taken on faith on a first reading.

Learning objectives. After completing this lesson, you will be able to:

- **Define** a normal-form game, strategy profile, payoff function, and bimatrix representation.
- **Classify** strategies as strictly dominant, strictly dominated, or neither, and apply iterated elimination of strictly dominated strategies (IESDS).
- **Compute** best response correspondences for any finite bimatrix game and use them to find all pure Nash equilibria.
- **Prove** that dominant strategy equilibria are unique Nash equilibria and that every Nash equilibrium survives IESDS.
- **Analyze** the Prisoner's Dilemma, Battle of the Sexes, Matching Pennies, and Stag Hunt, explaining why each exhibits qualitatively different equilibrium behavior.
- **Explain** the relationship between IESDS survival and rationalizability in two-player games.

2 References

- Osborne and Rubinstein [1994, Chapters 1–3] — Comprehensive introduction to normal-

form games, dominance, and Nash equilibrium with careful proofs.
- Fudenberg and Tirole [1991, Chapters 1–2] — Graduate-level treatment of strategic-form games, rationalizability, and iterated dominance.
- Shoham and Leyton-Brown [2008, Chapters 3–4] — Game theory from the perspective of multiagent systems, with connections to AI and MARL.
- Nash [1950] — Nash's original one-page proof of equilibrium existence (uses mixed strategies; covered in Lesson 34).

3 Games as Models of Strategic Interaction

Before introducing formal definitions, we describe the three ingredients that make up every game.

Players. A game involves two or more *players* (also called *agents* in the MARL literature). Each player is a distinct decision-maker with their own objectives. In the ride-sharing example, the two companies are the players.

Strategies. Each player chooses from a set of available actions, called *strategies*. Crucially, players choose *simultaneously*: no one observes the others' choices before committing to their own. Even if decisions are not literally simultaneous in clock time, what matters is that each player acts without knowing the others' choices.

Payoffs. Once all players have chosen their strategies, each player receives a numerical *payoff*. The payoff depends on *everyone's* choice, not just one's own. This is the key difference from single-agent optimization: your outcome is determined jointly by your action and the actions of others.

Reading a payoff table. For two-player games with finitely many strategies, we organize payoffs into a table. Each row corresponds to a strategy of player 1 (the "row player") and each column to a strategy of player 2 (the "column player"). Each cell contains a pair (a, b): the first number a is player 1's payoff, and the second b is player 2's payoff.

For instance, consider a simplified pricing game where each company chooses *High* (H) or *Low* (L):

	H	L
H	$(5, 5)$	$(1, 6)$
L	$(6, 1)$	$(2, 2)$

If company 1 chooses High and company 2 chooses Low, the entry is $(1, 6)$: company 1 earns 1 (losing most of its customers) while company 2 earns 6 (capturing the market). Notice that Low gives a higher payoff regardless of the opponent's choice ($6 > 5$ and $2 > 1$), so Low is the "obvious" strategy—a situation we will formalize as *strict dominance*.

With these three ingredients in mind, we now give precise definitions.

4 Normal-Form Games

Definition 4.1 (Normal-Form Game)**.** A *normal-form game* (or *strategic-form game*) is a tuple $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ where:

(i) $N = \{1, 2, \ldots, n\}$ is a finite set of *players*.

(ii) For each $i \in N$, S_i is a non-empty set of *strategies* (or *actions*) available to player i. The set of *strategy profiles* is $S = \prod_{i \in N} S_i$.

(iii) For each $i \in N$, $u_i \colon S \to \mathbb{R}$ is player i's *payoff function* (or *utility function*).

The game is *finite* if every S_i is a finite set.

Remark 4.2 (Notation). For a strategy profile $s = (s_1, \ldots, s_n) \in S$, we write $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n) \in S_{-i} := \prod_{j \neq i} S_j$ and use $s = (s_i, s_{-i})$ to emphasize player i's component. This notation separates "my choice" (s_i) from "everyone else's choices" (s_{-i}) and is used throughout game theory and MARL.

Definition 4.3 (Bimatrix Game)**.** A *bimatrix game* is a two-player finite normal-form game presented as a pair of matrices $(A, B) \in \mathbb{R}^{m \times k} \times \mathbb{R}^{m \times k}$, where player 1 has m strategies, player 2 has k strategies, $A_{jl} = u_1(j, l)$, and $B_{jl} = u_2(j, l)$. If $B = -A$, the game is *zero-sum*: whatever one player gains, the other loses.

We now introduce four classic two-player games that illustrate qualitatively different strategic situations. These games will serve as running examples throughout the lesson.

Example 4.4 (Prisoner's Dilemma)**.** Two suspects are interrogated separately. Each can *Cooperate* (C, stay silent) or *Defect* (D, betray the other). The payoff bimatrix is:

	C	D
C	$(3, 3)$	$(0, 5)$
D	$(5, 0)$	$(1, 1)$

Each entry (a, b) gives (row player's payoff, column player's payoff). Regardless of the opponent's choice, Defect yields a strictly higher payoff: if the opponent cooperates, D gives 5 versus C's 3; if the opponent defects, D gives 1 versus C's 0. This makes D a *strictly dominant strategy* for both players—a concept we formalize in Section 5.

Example 4.5 (Battle of the Sexes)**.** Players 1 and 2 prefer different activities—*Opera* (O) and *Football* (F)—but both prefer coordination to miscoordination:

	O	F
O	$(3, 1)$	$(0, 0)$
F	$(0, 0)$	$(1, 3)$

Neither player has a dominant strategy: player 1 prefers O if player 2 chooses O, but prefers F if player 2 chooses F. The "right" choice depends entirely on what the other player does. Both (O, O) and (F, F) will turn out to be pure Nash equilibria (Section 7).

Example 4.6 (Matching Pennies)**.** A zero-sum game: each player shows *Heads* (H) or *Tails* (T). Player 1 wins if the choices match; player 2 wins if they differ:

	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

Player 1 wants to match; player 2 wants to mismatch. No matter what profile is played, one player wants to switch. This game has no pure Nash equilibrium (Proposition 8.5), motivating the extension to mixed strategies in Lesson 34.

Example 4.7 (Stag Hunt)**.** Two hunters choose to hunt a *Stag* (S, requires cooperation) or a *Hare* (H, safe but less rewarding):

	S	H
S	$(4, 4)$	$(0, 3)$
H	$(3, 0)$	$(3, 3)$

Both (S, S) and (H, H) are pure Nash equilibria. The equilibrium (S, S) is *payoff-dominant* (higher payoffs for both players), while (H, H) is *risk-dominant* (safer against uncertainty about the opponent's choice). This tension between ambition and safety recurs throughout cooperative MARL.

RL/ML Connection

The Stag Hunt captures a fundamental tension in cooperative MARL: agents can achieve higher collective reward through coordination, but each agent bears risk if the others fail to coordinate. This is precisely the challenge faced by centralized-training decentralized-execution (CTDE) algorithms such as QMIX and MAPPO, where the training procedure must guide agents toward the payoff-dominant equilibrium despite the safety of the risk-dominant one.

These four games illustrate four qualitatively different situations: a game with a dominant strategy (Prisoner's Dilemma), a coordination game with multiple equilibria (Battle of the Sexes), a purely competitive game with no pure equilibrium (Matching Pennies), and a game where equilibrium selection is the central challenge (Stag Hunt). The rest of this lesson develops the tools to analyze each systematically.

5 Dominance and Iterated Elimination

The simplest way to reason about a game is to ask: are there strategies that a rational player would *never* use? If one strategy gives strictly less payoff than another *no matter what the opponents do*, then there is no reason to play it. This idea leads to the concept of dominance.

Definition 5.1 (Strict and Weak Dominance)**.** Let $\Gamma = (N, (S_i), (u_i))$ be a normal-form game and fix player i.

(a) A strategy $s_i \in S_i$ *strictly dominates* $s_i' \in S_i$ if

$$u_i(s_i, s_{-i}) > u_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

In words: s_i gives a strictly higher payoff than s_i' against *every possible* combination of opponents' strategies.

(b) A strategy s_i *weakly dominates* s_i' if

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i},$$

with strict inequality for at least one s_{-i}.

A strategy is *strictly dominated* if some other strategy strictly dominates it. A strategy is a *strictly dominant strategy* if it strictly dominates every other strategy of that player.

Example 5.2 (Dominance in the Prisoner's Dilemma)**.** In Example 4.4, consider player 1. Compare D against C for each possible choice by player 2:

- If player 2 plays C: $u_1(D, C) = 5 > 3 = u_1(C, C)$.
- If player 2 plays D: $u_1(D, D) = 1 > 0 = u_1(C, D)$.

Since D gives a higher payoff than C in *both* cases, D strictly dominates C for player 1. By the symmetry of the payoff matrix, D also strictly dominates C for player 2. Both players have a strictly dominant strategy.

Proposition 5.3 (Dominated Strategies Are Never Best Responses)**.** *If s_i' is strictly dominated by s_i, then s_i' is never a best response: for every $s_{-i} \in S_{-i}$, $s_i' \notin \arg\max_{t_i \in S_i} u_i(t_i, s_{-i})$.*

Proof. Fix any $s_{-i} \in S_{-i}$. Since s_i strictly dominates s_i', we have $u_i(s_i, s_{-i}) > u_i(s_i', s_{-i})$. Therefore s_i' does not achieve the maximum of $u_i(\cdot, s_{-i})$ over S_i, so $s_i' \notin \arg\max_{t_i \in S_i} u_i(t_i, s_{-i})$. □

Warning

The converse of Proposition 5.3 is false in general for pure strategies: a strategy can fail to be a best response to any pure opponent profile without being strictly dominated by any single pure strategy. However, in two-player games with finite strategy sets, a pure strategy that is never a best response is strictly dominated by some *mixed* strategy (a consequence of the supporting hyperplane theorem).

Since dominated strategies are never rational to play, we can remove them from consideration. But removing one player's dominated strategy may cause a previously non-dominated strategy of another player to become dominated. This motivates iterating the elimination process.

Definition 5.4 (Iterated Elimination of Strictly Dominated Strategies)**.** *Iterated elimination of strictly dominated strategies* (IESDS) proceeds as follows. Set $S_i^0 = S_i$ for all $i \in N$. For $k = 0, 1, 2, \ldots$, define

$$S_i^{k+1} = \{s_i \in S_i^k : \text{no } s_i' \in S_i^k \text{ satisfies } u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}) \text{ for all } s_{-i} \in S_{-i}^k\}.$$

The set of *surviving strategies* for player i is $S_i^\infty = \bigcap_{k=0}^\infty S_i^k$. In a finite game, this intersection stabilizes after finitely many rounds.

Remark 5.5. Definition 5.4 eliminates *all* strictly dominated strategies simultaneously at each round. One could instead eliminate strategies one at a time, choosing different players and strategies in different orders. Theorem 5.8 shows that the final surviving set is the same regardless of the elimination order.

Example 5.6 (IESDS in the Prisoner's Dilemma)**.** In the Prisoner's Dilemma (Example 4.4), C is strictly dominated by D for both players. Round 0 eliminates C for both, leaving $S_1^1 = S_2^1 = \{D\}$. The unique surviving profile is (D, D).

Example 5.7 (Multi-Round IESDS)**.** Consider the game:

	L	C	R
T	$(1,2)$	$(2,1)$	$(5,0)$
B	$(4,1)$	$(3,5)$	$(6,2)$

Round 0. Compare player 1's strategies: $u_1(B, s_2) > u_1(T, s_2)$ for every s_2 (specifically $4 > 1$, $3 > 2$, $6 > 5$). So T is strictly dominated by B. Eliminate T; the reduced game is:

	L	C	R
B	$(4,1)$	$(3,5)$	$(6,2)$

Round 1. Player 2 now knows player 1 will play B. Compare player 2's payoffs: L gives 1, C gives 5, R gives 2. Here C strictly dominates both L and R. Eliminate L and R.

The surviving profile is (B, C) with payoffs $(3, 5)$. Notice that C did *not* dominate L in the original game (against T, L gives 2 while C gives 1); the domination only emerged after eliminating player 1's dominated strategy.

Theorem 5.8 (Order Independence of IESDS)**.** *In a finite normal-form game, the set of strategy profiles surviving IESDS is independent of the order in which strategies are eliminated.*

Proof. We show that any strategy eliminated under one elimination order is eventually eliminated under every elimination order. The proof proceeds by strong induction on the round of elimination.

Let E and E' denote two arbitrary elimination orders. Write $S_i^k(E)$ for the surviving strategies of player i after round k under order E, and similarly $S_i^k(E')$ for order E'. Write $S_i^\infty(E)$ and $S_i^\infty(E')$ for the respective final surviving sets.

We prove by strong induction on k that if a strategy s_i^* is eliminated at round k under order E, then $s_i^* \notin S_i^\infty(E')$.

Base case ($k = 0$). If s_i^* is eliminated at round 0 under E, then there exists $s_i' \in S_i$ with $u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$ for all $s_{-i} \in S_{-i}$. Since $S_{-i}^\infty(E') \subseteq S_{-i}$, the domination holds on $S_{-i}^\infty(E')$. If $s_i' \in S_i^\infty(E')$, then $s_i^* \notin S_i^\infty(E')$ directly. Otherwise, following the chain of dominators from s_i' to some $\hat{s}_i \in S_i^\infty(E')$ (the chain terminates since S_i is finite), we get $u_i(\hat{s}_i, s_{-i}) > u_i(s_i^*, s_{-i})$ on $S_{-i}^\infty(E')$, so again $s_i^* \notin S_i^\infty(E')$.

Inductive step. Assume the claim holds for all strategies eliminated in rounds $0, 1, \ldots, k-1$ under any elimination order. Suppose s_i^* is eliminated at round k under order E. Then there exists $s_i' \in S_i^{k-1}(E)$ such that

$$u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}^{k-1}(E). \tag{1}$$

Claim: $S_{-i}^\infty(E') \subseteq S_{-i}^{k-1}(E)$.

To prove the claim, take any $s_j \in S_j^\infty(E')$ for $j \neq i$. Suppose for contradiction that $s_j \notin S_j^{k-1}(E)$. Then s_j was eliminated at some round $l < k$ under order E. By the inductive hypothesis, $s_j \notin S_j^\infty(E')$, contradicting $s_j \in S_j^\infty(E')$. Hence $s_j \in S_j^{k-1}(E)$, establishing the claim.

By the claim, the strict inequality (1) holds for all $s_{-i} \in S_{-i}^\infty(E')$. We now consider two cases.

Case 1: $s_i' \in S_i^\infty(E')$. Then s_i^* is strictly dominated by s_i' on the surviving set $S_{-i}^\infty(E')$, so $s_i^* \notin S_i^\infty(E')$.

Case 2: $s_i' \notin S_i^\infty(E')$. Then s_i' was eliminated at some round m under E', meaning there exists $s_i'' \in S_i^{m-1}(E')$ that strictly dominates s_i' on $S_{-i}^{m-1}(E')$. Repeating this chain, since S_i is finite, we eventually reach some $\hat{s}_i \in S_i^\infty(E')$ that strictly dominates s_i' on $S_{-i}^\infty(E')$. Combining with (1): for all $s_{-i} \in S_{-i}^\infty(E')$, $u_i(\hat{s}_i, s_{-i}) > u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$. Hence s_i^* is strictly dominated by $\hat{s}_i$ on the surviving set, so $s_i^* \notin S_i^\infty(E')$.

By induction, $S_i^\infty(E) \supseteq S_i^\infty(E')$ for all i. Swapping the roles of E and E' gives the reverse inclusion, so $S_i^\infty(E) = S_i^\infty(E')$. □

Key Result

The surviving set under IESDS is a canonical object of the game: it does not depend on procedural choices. In a game solvable by IESDS (where the surviving set is a singleton), IESDS yields the unique Nash equilibrium.

6 Best Response and Rationalizability

Dominance asks "what should I never play?" We now ask the complementary question: "what *should* I play?" The answer depends on what I expect my opponents to do. Given a belief about the opponents' strategies, the rational choice is whichever strategy maximizes my payoff—this is called a *best response*.

Definition 6.1 (Best Response Correspondence)**.** Player i's *best response correspondence* is the set-valued map $\mathrm{BR}_i \colon S_{-i} \rightrightarrows S_i$ defined by

$$\mathrm{BR}_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

In a finite game, $\mathrm{BR}_i(s_{-i})$ is non-empty for every $s_{-i} \in S_{-i}$, since a finite set always attains its maximum.

Example 6.2 (Best Responses in Battle of the Sexes)**.** In Example 4.5, compute player 1's best responses:

- If player 2 plays O: $u_1(O, O) = 3 > 0 = u_1(F, O)$, so $\mathrm{BR}_1(O) = \{O\}$.
- If player 2 plays F: $u_1(F, F) = 1 > 0 = u_1(O, F)$, so $\mathrm{BR}_1(F) = \{F\}$.

By symmetry (with swapped preferences), $\mathrm{BR}_2(O) = \{O\}$ and $\mathrm{BR}_2(F) = \{F\}$. The profiles where both players play a mutual best response are (O, O) and (F, F)—these are exactly the Nash equilibria, as we will see in Section 7.

Definition 6.3 (Rationalizable Strategies)**.** A strategy s_i is *rationalizable* if it survives iterated elimination of strategies that are never best responses. Formally, set $R_i^0 = S_i$ and define

$$R_i^{k+1} = \left\{ s_i \in R_i^k : \exists\, s_{-i} \in R_{-i}^k \text{ such that } s_i \in \mathrm{BR}_i(s_{-i}) \right\}.$$

The set of *rationalizable strategies* is $R_i^\infty = \bigcap_{k=0}^\infty R_i^k$.

Remark 6.4. Rationalizability captures "common knowledge of rationality": a strategy is rationalizable if it is a best response to some belief about the opponents, where that belief is consistent with the opponents being rational, and so on ad infinitum.

Proposition 6.5 (IESDS and Rationalizability in Two-Player Games)**.** *In a finite two-player game, the set of rationalizable strategies coincides with the set of strategies surviving IESDS: $R_i^\infty = S_i^\infty$ for $i = 1, 2$.*

Proof. We establish both inclusions.

($S_i^\infty \supseteq R_i^\infty$). Every strictly dominated strategy is never a best response (Proposition 5.3). Hence every strategy eliminated in IESDS is also eliminated in the rationalizability procedure, giving $R_i^k \subseteq S_i^k$ for all k by induction. Taking intersections, $R_i^\infty \subseteq S_i^\infty$.

($S_i^\infty \subseteq R_i^\infty$). We show the contrapositive: if $s_i \notin R_i^\infty$, then $s_i \notin S_i^\infty$. Suppose s_i is eliminated at round k in the rationalizability procedure, meaning s_i is not a best response to any $s_{-i} \in R_{-i}^{k-1}$.

In a two-player game, this means: for every $s_j \in R_j^{k-1}$ (where $j \neq i$), there exists $s_i'(s_j) \in R_i^{k-1}$ with $u_i(s_i'(s_j), s_j) > u_i(s_i, s_j)$. We claim that s_i is strictly dominated on R_j^{k-1} (possibly by a mixed strategy). Indeed, consider the payoff vectors $v = (u_i(s_i, s_j))_{s_j \in R_j^{k-1}} \in \mathbb{R}^{|R_j^{k-1}|}$ and the convex set $W = \{(u_i(s_i', s_j))_{s_j \in R_j^{k-1}} : s_i' \in R_i^{k-1}\}$. Since s_i is never a best response on R_j^{k-1}, by the supporting hyperplane theorem applied to the convex hull of payoff vectors, there exists a convex combination (i.e., a mixed strategy) over R_i^{k-1} that strictly dominates s_i on R_j^{k-1}.

For pure-strategy IESDS where we also allow elimination by mixed-strategy domination, this gives $s_i \notin S_i^\infty$. In finite two-player games, elimination by pure or mixed strict domination yields the same surviving set [Osborne and Rubinstein, 1994], completing the proof. □

Warning

For games with three or more players, the equivalence between IESDS and rationalizability can fail when restricting to pure-strategy domination. Rationalizability may be strictly coarser than pure-strategy IESDS survival for $n \geq 3$. The equivalence is restored when domination by mixed strategies is permitted [Fudenberg and Tirole, 1991].

7 Pure Strategy Nash Equilibrium

We arrive at the central concept of game theory. Informally, a *Nash equilibrium* is a strategy profile where every player is simultaneously doing the best they can, given what everyone else is doing. If the game were paused and each player were told the others' strategies, no one would want to change their own.

Definition 7.1 (Pure Strategy Nash Equilibrium)**.** A strategy profile $s^* = (s_1^*, \ldots, s_n^*) \in S$ is a *pure strategy Nash equilibrium* (NE) if no player can unilaterally improve their payoff:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for all } s_i \in S_i, \text{ for all } i \in N.$$

Proposition 7.2 (Nash Equilibrium as Mutual Best Response)**.** *A strategy profile s^* is a Nash equilibrium if and only if $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for every $i \in N$.*

Proof. By Definition 7.1, s^* is a NE if and only if $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$ for all $s_i \in S_i$ and all $i \in N$. By Definition 6.1, this is precisely $s_i^* \in \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}^*) = \mathrm{BR}_i(s_{-i}^*)$ for all i. □

Proposition 7.2 gives a practical algorithm for finding all pure Nash equilibria in a bimatrix game: for each cell in the payoff table, check whether *both* players are playing a best response.

Example 7.3 (The "Underline Best Responses" Method)**.** Consider the 3×2 game:

	L	R
T	$(2,1)$	$(0,3)$
M	$(3,0)$	$(1,2)$
B	$(1,2)$	$(2,1)$

Step 1: Find player 1's best responses (compare payoffs within each column).

- Column L: payoffs are $2, 3, 1$. Max is 3 at M, so $\mathrm{BR}_1(L) = \{M\}$.
- Column R: payoffs are $0, 1, 2$. Max is 2 at B, so $\mathrm{BR}_1(R) = \{B\}$.

Step 2: Find player 2's best responses (compare payoffs within each row).

- Row T: payoffs are $1, 3$. Max is 3 at R, so $\mathrm{BR}_2(T) = \{R\}$.
- Row M: payoffs are $0, 2$. Max is 2 at R, so $\mathrm{BR}_2(M) = \{R\}$.
- Row B: payoffs are $2, 1$. Max is 2 at L, so $\mathrm{BR}_2(B) = \{L\}$.

Step 3: Find mutual best responses. A cell is a NE if and only if both players' entries are best responses. Here M is player 1's best response to L, but player 2's best response to M is R (not L); and B is player 1's best response to R, but player 2's best response to B is L (not R). No cell has both entries as best responses, so this game has *no pure Nash equilibrium.*

Key Result

The "underline best responses" method from Example 7.3—mark the highest payoff in each column for player 1 and each row for player 2, then look for cells with both entries marked—works for any bimatrix game and is the standard way to find pure NE by hand.

Proposition 7.4 (Dominant Strategy Equilibrium Implies Nash Equilibrium)**.** *If s_i^* is a strictly dominant strategy for every player $i \in N$, then $(s_1^*, \ldots, s_n^*)$ is the unique Nash equilibrium.*

Proof. Since s_i^* strictly dominates every other strategy of player i, we have $u_i(s_i^*, s_{-i}) > u_i(s_i, s_{-i})$ for all $s_i \neq s_i^*$ and all $s_{-i} \in S_{-i}$. In particular, for $s_{-i} = s_{-i}^*$, this gives $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for all i, so s^* is a NE by Proposition 7.2.

For uniqueness, suppose $\hat{s} \neq s^*$ is a NE. Then some $\hat{s}_i \neq s_i^*$, but $u_i(s_i^*, \hat{s}_{-i}) > u_i(\hat{s}_i, \hat{s}_{-i})$ by strict dominance, contradicting $\hat{s}_i \in \mathrm{BR}_i(\hat{s}_{-i})$. □

8 Analysis of Classic Games

We systematically analyze the four classic games from Section 4, identifying all pure Nash equilibria and highlighting the qualitative differences between them.

8.1 Prisoner's Dilemma

Proposition 8.1. *The Prisoner's Dilemma (Example 4.4) has a unique pure Nash equilibrium at* (D, D) *with payoffs* $(1, 1)$.

Proof. By Example 5.2, D is a strictly dominant strategy for both players. By Proposition 7.4, (D, D) is the unique Nash equilibrium. □

Remark 8.2 (Individual vs. Collective Rationality). The profile (C, C) yields payoffs $(3, 3)$, which both players prefer to the NE payoff of $(1, 1)$. Yet rational players cannot sustain cooperation in a one-shot game. This tension between individual and collective rationality is central to MARL: independent learners converge to the suboptimal NE unless the interaction is repeated (Lesson 36, folk theorems) or agents can commit to binding agreements.

8.2 Battle of the Sexes

Proposition 8.3. *The Battle of the Sexes (Example 4.5) has exactly two pure Nash equilibria:* (O, O) *with payoffs* $(3, 1)$ *and* (F, F) *with payoffs* $(1, 3)$.

Proof. From Example 6.2, $\mathrm{BR}_1(O) = \{O\}$, $\mathrm{BR}_1(F) = \{F\}$, $\mathrm{BR}_2(O) = \{O\}$, and $\mathrm{BR}_2(F) = \{F\}$. Of the four profiles, only (O, O) and (F, F) satisfy $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for both players. At (O, F), player 1 deviates ($O \notin \mathrm{BR}_1(F)$); at (F, O), likewise. □

Remark 8.4 (Equilibrium Selection). The Battle of the Sexes illustrates the *equilibrium selection problem*: the theory predicts multiple equilibria but does not say which will be played. In MARL, independent learners may converge to different equilibria or cycle. Correlated equilibrium (Lesson 34) offers one way to resolve this ambiguity.

8.3 Matching Pennies

Proposition 8.5 (Matching Pennies Has No Pure NE)**.** *The Matching Pennies game (Example 4.6) has no pure strategy Nash equilibrium.*

Proof. Player 1 wants to match and player 2 wants to mismatch, so their best responses oppose each other: $\mathrm{BR}_1(H) = \{H\}$, $\mathrm{BR}_1(T) = \{T\}$, $\mathrm{BR}_2(H) = \{T\}$, $\mathrm{BR}_2(T) = \{H\}$. At every profile, exactly one player has a profitable deviation:

- (H, H): player 2 deviates ($H \notin \mathrm{BR}_2(H)$).
- (H, T): player 1 deviates ($H \notin \mathrm{BR}_1(T)$).
- (T, H): player 1 deviates ($T \notin \mathrm{BR}_1(H)$).
- (T, T): player 2 deviates ($T \notin \mathrm{BR}_2(T)$).

No profile is a mutual best response, so no pure NE exists. □

Remark 8.6. The non-existence of pure NE reflects the adversarial structure of zero-sum games. Nash's theorem [Nash, 1950] guarantees a *mixed* NE (each player randomizes uniformly), which is the subject of Lesson 34. In competitive MARL, the best-response cycling $H \to T \to H \to \cdots$ is a simple instance of the oscillation that plagues naive independent learning; algorithms like minimax-Q and self-play converge to mixed equilibria instead.

8.4 Stag Hunt

Proposition 8.7. *The Stag Hunt (Example 4.7) has exactly two pure Nash equilibria:* (S, S) *with payoffs* $(4, 4)$ *and* (H, H) *with payoffs* $(3, 3)$.

Proof. Best responses: $\mathrm{BR}_1(S) = \{S\}$ (since $4 > 3$), $\mathrm{BR}_1(H) = \{H\}$ (since $3 > 0$), and by symmetry $\mathrm{BR}_2(S) = \{S\}$, $\mathrm{BR}_2(H) = \{H\}$. Only the diagonal profiles (S, S) and (H, H) satisfy mutual best response; at (S, H) and (H, S), the Stag-hunter deviates. □

Remark 8.8 (Payoff Dominance vs. Risk Dominance). The equilibrium (S, S) is *payoff-dominant* (both players prefer it), while (H, H) is *risk-dominant* [Fudenberg and Tirole, 1991]: S is a best response only when the believed probability p that the opponent plays S satisfies $4p > 3p + 3(1 - p)$, i.e., $p > 3/4$. Unless a player is quite confident the other will cooperate, the safe choice is H. In MARL, this tension determines whether independent learners converge to the Pareto-optimal or the risk-dominant equilibrium.

9 Exercises

Exercise 9.1 (Three-Player Dominance)**.** $(*)$
Consider a three-player game where each player chooses from $\{A, B\}$. Player 1's payoff is $u_1(s_1, s_2, s_3) = \mathbf{1}[s_1 = A] \cdot (1 + \mathbf{1}[s_2 = A] + \mathbf{1}[s_3 = A])$. That is, player 1 gets 0 from B and gets 1, 2, or 3 from A depending on how many others play A.

(a) Show that B is strictly dominated by A for player 1.

(b) Suppose u_2 and u_3 are defined analogously. Find all pure Nash equilibria.

(c) Apply IESDS and verify that the surviving set equals the set of Nash equilibria.

Exercise 9.2 (Best Response Analysis)**.** $(*)$
Consider the bimatrix game:

	L	C	R
T	$(2, 1)$	$(0, 2)$	$(1, 0)$
M	$(1, 2)$	$(3, 1)$	$(0, 0)$
B	$(0, 0)$	$(1, 3)$	$(2, 1)$

(a) Compute $\mathrm{BR}_1(s_2)$ for each $s_2 \in \{L, C, R\}$ and $\mathrm{BR}_2(s_1)$ for each $s_1 \in \{T, M, B\}$.

(b) Identify all pure Nash equilibria using the mutual best response characterization (Proposition 7.2).

(c) Apply IESDS. Does IESDS alone determine the equilibrium?

Exercise 9.3 (IESDS Order Independence)**.** $(**)$
Consider the following 3×3 bimatrix game:

	L	C	R
T	$(3, 1)$	$(0, 2)$	$(1, 0)$
M	$(1, 0)$	$(2, 3)$	$(0, 1)$
B	$(0, 2)$	$(1, 1)$	$(3, 0)$

(a) Perform IESDS by first eliminating dominated strategies for player 1, then player 2.

(b) Perform IESDS in the reverse order: first player 2, then player 1.

(c) Verify that both orders yield the same surviving set, confirming Theorem 5.8.

Exercise 9.4 (Coordination Game Design). (∗∗)
Design a symmetric 3×3 bimatrix game (i.e., $B = A^T$) with exactly three pure NE (all on the diagonal) and no strictly dominated strategies. Prove both properties hold.

Exercise 9.5 (Nash Equilibrium and IESDS Survival). (∗∗)
Prove that every Nash equilibrium of a finite game survives IESDS: if s^* is a pure NE, then $s_i^* \in S_i^\infty$ for all $i \in N$.
Hint: Induction on the elimination round.

Exercise 9.6 (Equilibrium Selection in Multi-Agent Training). (∗ ∗ ∗)
Consider two Q-learning agents playing the Stag Hunt (Example 4.7) repeatedly with learning rate α_t and ε-greedy exploration. At each round, both agents observe the joint action and update their Q-tables independently.

(a) Suppose both agents initialize $Q(S) = Q(H) = 0$. If the first outcome is (H, H) with reward 3, show that both agents increase $Q(H)$ and are subsequently more likely to play H, making (H, H) self-reinforcing.

(b) Argue informally that for sufficiently small ε, the risk-dominant equilibrium (H, H) is more likely to emerge than the payoff-dominant (S, S). Why does the basin of attraction of (H, H) matter here?

(c) Suggest a modification to the training procedure (e.g., optimistic initialization, reward shaping, or communication) that could steer convergence toward the payoff-dominant equilibrium. Justify your proposal using the best-response structure of the game.

10 Summary

- A **normal-form game** $(N, (S_i), (u_i))$ models simultaneous strategic interaction: each player's payoff depends on everyone's choice, not just their own.
- A strategy is **strictly dominated** if another strategy gives a strictly higher payoff against every possible opponent behavior; rational players never use dominated strategies.
- **IESDS** iteratively removes dominated strategies. The surviving set is independent of elimination order (Theorem 5.8) and is a canonical property of the game.
- A **best response** is a strategy that maximizes payoff given the opponents' choices. The best response correspondence BR_i maps each opponent profile to the set of optimal replies.
- A **pure Nash equilibrium** is a profile where every player simultaneously plays a best response (Proposition 7.2); no one can unilaterally improve.
- If every player has a strictly dominant strategy, the resulting profile is the unique NE (Proposition 7.4).
- Not every game has a pure NE (Matching Pennies); some have multiple NE (Battle of the Sexes, Stag Hunt), raising the **equilibrium selection problem**.
- In two-player games, the set of rationalizable strategies equals the IESDS surviving set (Proposition 6.5).
- These concepts underpin every MARL algorithm: convergence targets, solution quality, and equilibrium selection all rely on game-theoretic reasoning developed here.

11 Further Reading

- **Lesson 34** extends the analysis to *mixed strategies* and proves Nash's existence theorem: every finite game has at least one (possibly mixed) Nash equilibrium. The minimax theorem for zero-sum games is also covered there.
- **Lesson 35** introduces *extensive-form games* (sequential decision trees), perfect information, and subgame perfect equilibrium—the appropriate solution concept when players move in sequence rather than simultaneously.
- **Lesson 36** treats *repeated games* and folk theorems, explaining how cooperation can emerge when the Prisoner's Dilemma is played repeatedly with patient agents.
- For a deeper treatment of rationalizability and epistemic foundations, see Osborne and Rubinstein [1994, Chapters 4–5] and Fudenberg and Tirole [1991, Chapter 2].
- von Neumann and Morgenstern [1944] is the foundational work that launched game theory as a mathematical discipline; Nash [1951] introduced the equilibrium concept that bears his name.

References

Drew Fudenberg and Jean Tirole. *Game Theory*. MIT Press, 1991.

John F. Nash. Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences*, 36(1):48–49, 1950.

John F. Nash. Non-cooperative games. *Annals of Mathematics*, 54(2):286–295, 1951.

Martin J. Osborne and Ariel Rubinstein. *A Course in Game Theory*. MIT Press, 1994.

Yoav Shoham and Kevin Leyton-Brown. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press, 2008.

John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, 1944.